

Antonio De Leon

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Summary

Ph.D. candidate in Statistics building Bayesian forecasting, quantile-modeling, and interval-methods tools for environmental and financial time series. I work across model development, R/Python software, diagnostics, and reproducible compute workflows.

Research Software

exdqlm

CRAN v1.1.0; JSS software article submitted

R package for dynamic and static Bayesian quantile models under the exAL family. Co-author of software supporting dynamic exDQLMs, static exAL regression, MCMC, Laplace–delta variational Bayes, diagnostics, and posterior predictive synthesis.

Education

Ph.D. in Statistics, University of California, Santa Cruz

Expected Aug. 2026

Research: Bayesian time series, quantile modeling, forecast correction and synthesis, scalable variational Bayes inference.

M.Sc. in Economics, Instituto Tecnológico Autónomo de México (ITAM)

2018–2020

Mexico City, Mexico

B.Sc. in Applied Mathematics, Instituto Tecnológico Autónomo de México (ITAM)

2014–2018

Mexico City, Mexico

Selected Professional Experience

Delos Financial Technologies, San Rafael, CA (Remote)

May 2025–Sep. 2025

Quantitative Researcher (Contract)

Python, Git, Bash, C++, AWS, Stan

- Built model-evaluation notebooks for residual checks, goodness-of-fit plots, cohort summaries, and backtest review.
- Implemented baseline and extended forecasting variants, automated AWS experiment runs, and standardized versioned outputs for reproducible comparisons.

UCSC Statistics Department, Santa Cruz, CA

Feb. 2024–Present

Computer Systems Coordinator

Linux, Bash, Python/R, Networking, User Support

- Administer Linux research servers used by faculty and graduate students, including software stacks, storage, upgrades, and incident response.
- Wrote automation scripts and documentation for onboarding, resource usage, and reproducible compute workflows; coordinate with central IT on infrastructure planning.

NeatLeaf Inc., Scotts Valley, CA

May 2021–Sep. 2022

Data Analyst

Python, SQL, Git, Bash, Linux, C++, DBeaver

- Built Python/SQL pipelines for greenhouse robot and sensor telemetry used in monitoring and downstream analytics.
- Modeled spatiotemporal microclimate anomalies and analyzed robot/sensor calibration logs to improve data quality and system reliability.

Banco de México (BANXICO), Mexico City, Mexico

Aug. 2018–Dec. 2019

Data Analyst

Python, R, SQL, Git

- Built Python/R/SQL pipelines for banknote image datasets used in operational monitoring and model development.
- Developed anomaly-classification diagnostics and financial-volume forecasting prototypes for operational planning.

Selected Publications & Presentations

De Leon, A., Barata, R., Prado, R., & Sansó, B.

exdqlm: An R Package for Estimation and Analysis of Flexible Dynamic Quantile Linear Models.

Software article submitted to the *Journal of Statistical Software*; R package on CRAN v1.1.0; arXiv:2607.22760.

De Leon, A., Prado, R., & Sansó, B.

Bayesian Quantile Readouts for Fixed Deep Echo State Networks.

Working paper in progress on Bayesian quantile forecasting with fixed reservoir features, shrinkage priors, MCMC, and VB–LD.

De Leon, A., Prado, R., & Sansó, B.
Mean-Tilted Relaxed Quantile Regression: Fixed-Content Interval Functionals and Generalized-Bayes Computation.
arXiv preprint on generalized-Bayes computation for fixed-content interval-root functionals; arXiv:2607.26098.

De Leon, A., Sansó, B., & Prado, R.
Bayesian Quantile-Based Correction and Synthesis of Hydrologic Products.
Manuscript in revision at *Environmetrics*; related work received the ISBA 2026 Best Poster Prize and EnviBayes Student Paper Competition award.

De Leon, A., & Lobato, I. N.
Evidence of non-fundamentalness in OECD capital stocks.
Empirical Economics 67, 761–772 (2024). doi:10.1007/s00181-024-02564-5

ISBA World Meeting 2026, Nagoya, Japan
Poster presentation: “Bayesian Quantile-Based Correction and Synthesis of Climate Products.”
Awarded Best Poster Prize; presented July 1, 2026.

LACSC-TIES-EnviBayes-EnvrASA 2026, Mexico City, Mexico
Invited-session abstract submitted as a 2026 EnviBayes Student Paper Competition winner.
EnviBayes competition-winners invited session.

Awards, Fellowships & Competitive Funding

- **Best Poster Prize**, ISBA 2026 World Meeting, International Society for Bayesian Analysis, 2026.
- **ISBA 2026 Junior Researcher Travel Award**, International Society for Bayesian Analysis / National Science Foundation, 2026.
- **EnviBayes Student Paper Competition Winner**, International Society for Bayesian Analysis, 2026.
- **UCSC Statistics Department Conference Travel Award**, 2026; **UCSC Summer Research Fellowship**, 2024.
- **UC Regents Fellowship**, UCSC Statistics, 2021.
- **TLC Graduate Pedagogy Fellowship Certificate**, Center for Innovations in Teaching and Learning, 2023–2024.
- **Baillères Fellowship**, ITAM B.Sc. and M.Sc., 2014–2020.

Teaching & Mentoring

University of California, Santa Cruz	2021–Present
<i>Teaching Assistant / Graduate Student Instructor</i>	<i>Statistics and Probability</i>

- Graduate Student Instructor for *Data Visualization (STAT 80B)*, Spring 2025.
- Teaching Assistant for *Probability Theory (STAT 131)*, *Classical and Bayesian Inference (STAT 132)*, *Statistics (STAT 5)*, and related courses.

ASA DataFest, Fresno, CA	Apr. 2023
<i>Mentor</i>	<i>R, Python</i>

- Mentored student teams on statistical modeling, visualization, and technical communication.

Technical Skills

Programming	R, Python, SQL, C/C++, Bash, Git, $\LaTeX$
Methods	Bayesian inference, state-space models, time series forecasting, quantile modeling, variational inference, diagnostics, calibration
Tools	Linux, AWS (EC2, EMR, S3), Stan, Rcpp, CRAN, GitHub Actions, Tableau
Training	SISG, SISMID, causal inference with R
Languages	English, Spanish, German